

Combinational Logic Design Exercises 2, 3 and 4

RAHUL B 230508V

July 18, 2026

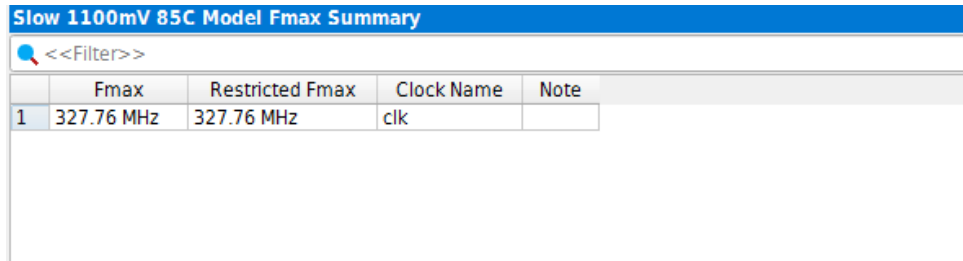
1 Performance and Resource Utilization Results

The results are retrieved by simulating the SystemVerilog files on Quartus Prime and using Cyclone V FPGA as the test device.

Architecture	Logic Utilization (ALMs)	Total Registers	Fmax (MHz)	Total Pins
Cascade Adder	19	0	-	98
Pipeline Adder	49	180	327.76	99
CLA (32-bit)	60	0	-	98
IEEE 754 FP Adder	324	0	-	96

Table 1: Comparative resource utilization and timing summary.

1.1 Synthesis Reports



Slow 1100mV 85C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	327.76 MHz	327.76 MHz	clk	

Figure 1: Timing analysis summary for the Pipelined Adder.

Fitter Summary	
<<Filter>>	
Fitter Status	Successful - Sat Jul 18 08:34:31 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	cascade_adder
Top-level Entity Name	pipelined_adder
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	49 / 56,480 (< 1 %)
Total registers	180
Total pins	99 / 268 (37 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total RAM Blocks	0 / 686 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 2: Logic utilization for the Pipelined Adder.

mantissa_adder.v	
normalization.v	
leading_zero_detector.v	
adder.v	
Compilation Report - adder	
Flow Summary	
<<Filter>>	
Flow Status	Successful - Sat Jul 18 10:42:29 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	adder
Top-level Entity Name	adder
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	324 / 56,480 (< 1 %)
Total registers	0
Total pins	96 / 268 (36 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 3: Logic utilization for the IEEE 754 Floating-Point Adder.

Fitter Summary	
<<Filter>>	
Fitter Status	Successful - Sat Jul 18 09:25:51 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	cascade_adder
Top-level Entity Name	cla_32bit
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	60 / 56,480 (< 1 %)
Total registers	0
Total pins	98 / 268 (37 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total RAM Blocks	0 / 686 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 4: Logic utilization for the 32-bit Carry Lookahead Adder (CLA).

Fitter Summary	
<<Filter>>	
Fitter Status	Successful - Sat Jul 18 08:38:34 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	cascade_adder
Top-level Entity Name	cascade_adder
Family	Cyclone V
Device	5CGXFC7C7F23C8
Timing Models	Final
Logic utilization (in ALMs)	19 / 56,480 (< 1 %)
Total registers	0
Total pins	98 / 268 (37 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total RAM Blocks	0 / 686 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 13 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 5: Logic utilization for the Cascaded Adder.